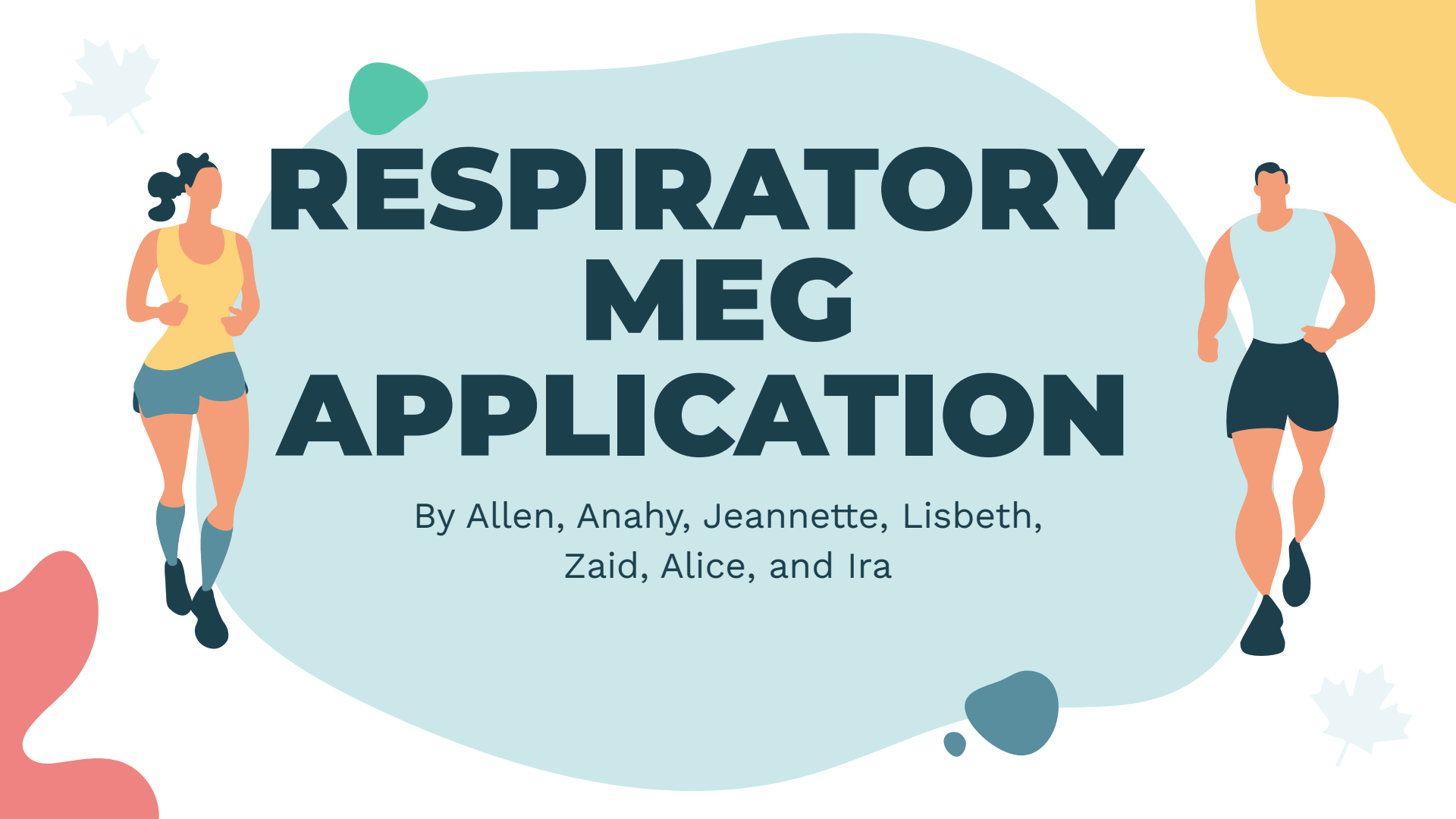


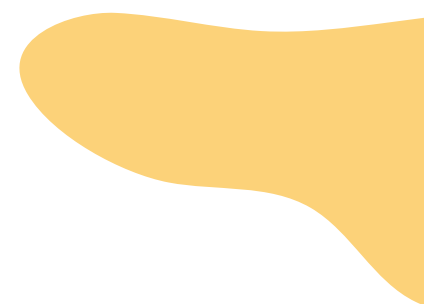


RESPIRATORY MEG APPLICATION

By Allen, Anahy, Jeannette, Lisbeth,
Zaid, Alice, and Ira



OUTLINE



01

MEG Application



02

Theory of MEG

How do MEGs work?



03

MEG Design

Design and How it will work



04

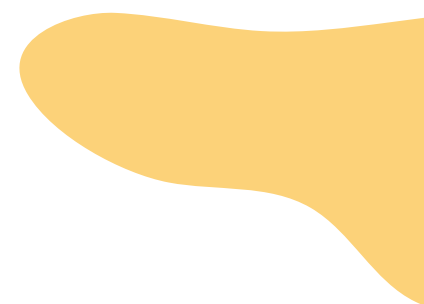
MEG Materials

Ecoflex, Magnetic Powder,
3D Mold, Copper Wire





OUTLINE



05

**Testing Protocol
& Data**



06

**Project Timeline &
Challenges**



07

Future Applications



08

Conclusion



01 MEG Application



THE BIG IDEA

- Build an MEG device that can track respiratory outputs.
- Tracking breathing patterns is important and can provide very useful data!
 - Health Monitoring
 - Biofeedback and Therapeutic Applications
 - Sport and Fitness Tracking
 - Sleep Analysis
 - Medical Research
- Additional Application:
 - Divers!



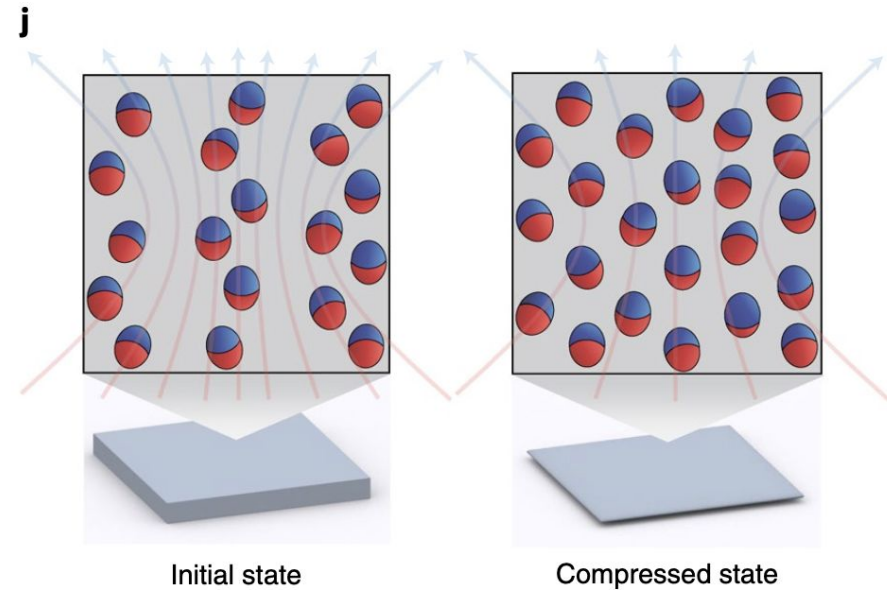
02

Theory of MEG



Theory of MEGs

- Relies on **two** main principles:
 - Magnetoelastic effect
 - Magnetic properties of a material change when subjected to mechanical stress
 - Electromagnetic induction
 - The production of current as a result of changing magnetic fields through a solenoid
- Giant magnetoelastic effect in soft systems



03

MEG Design

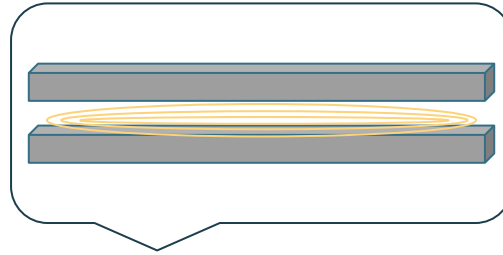


Compression Shirt Design

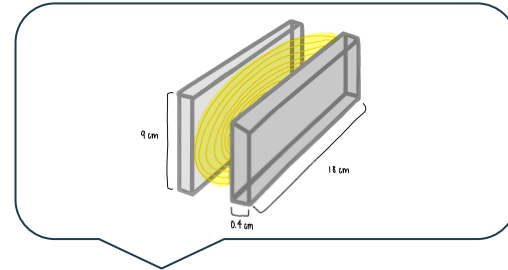


Shirt w/ MEG

MEG will be sewed onto the inside of the compression shirt

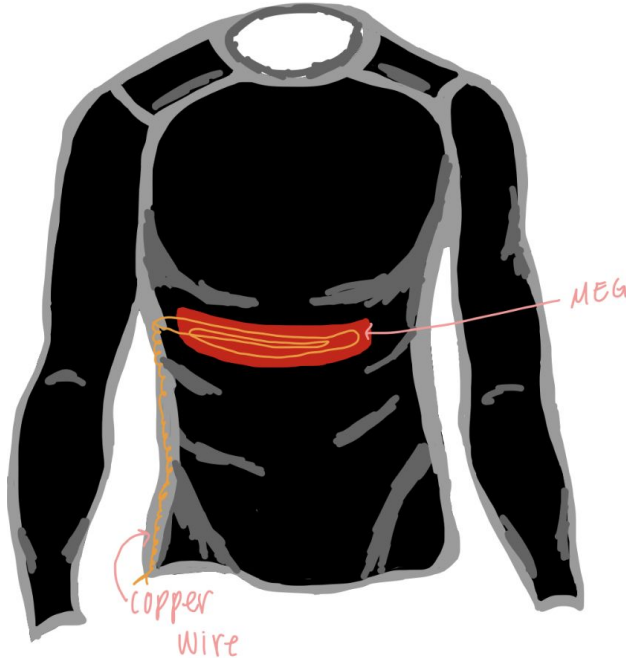


Aerial View



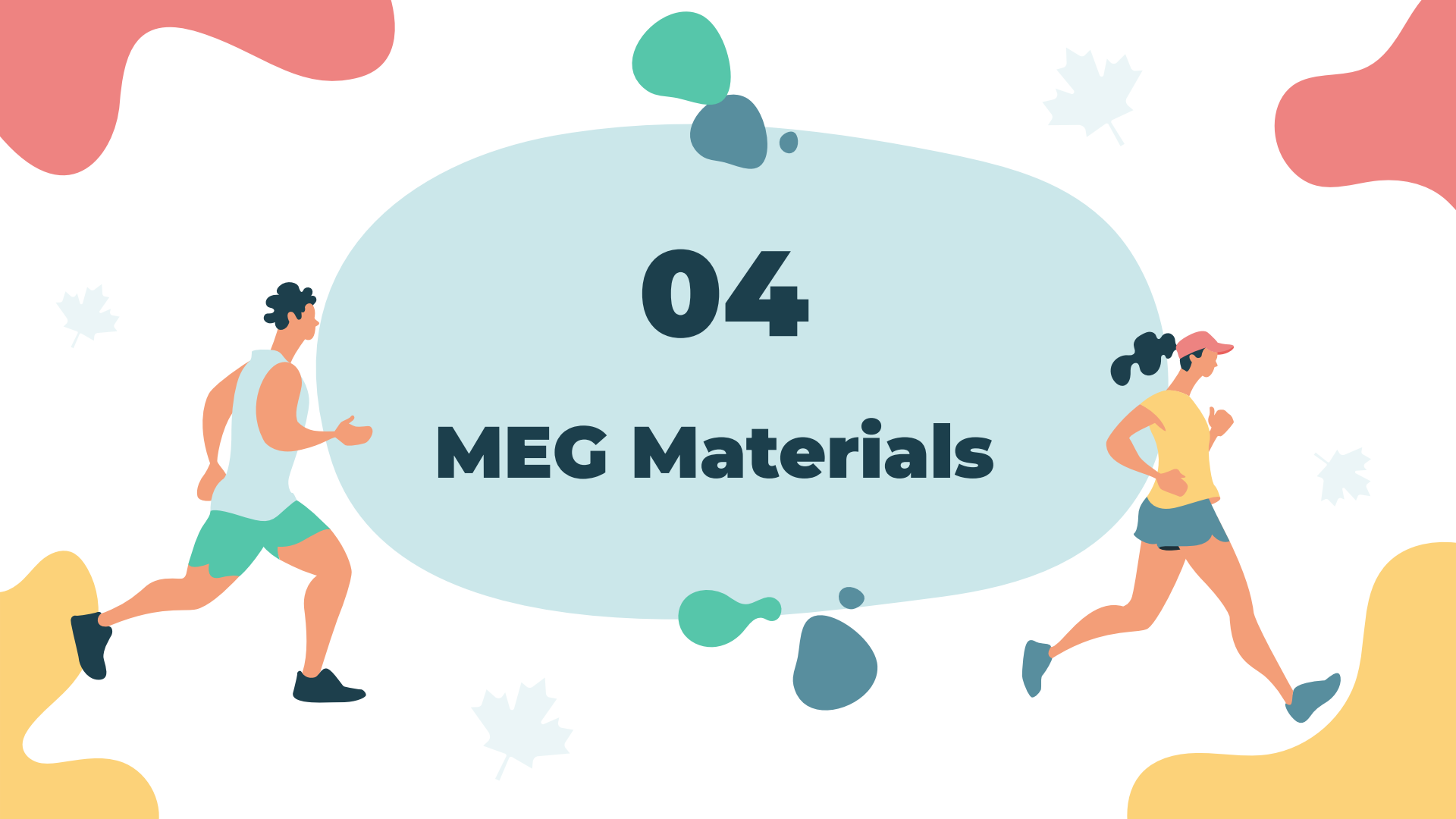
Side Exterior View

Compression Shirt Design



Chest movement will deform the MEGs and cause magnetic field distortion, inducing an electromotive force and generating electrical signals in the conductive coil

PLACED ON THE DIAPHRAGM

An illustration featuring two runners, a man on the left and a woman on the right, both in motion. The man is wearing a light blue tank top and green shorts, while the woman is wearing a yellow tank top, blue shorts, and a pink cap. They are flanking a large, light blue oval shape that contains the text. The background is white with stylized red and yellow shapes at the corners, green and blue abstract shapes around the central oval, and several light blue maple leaves scattered throughout.

04

MEG Materials

Materials



Ecoflex
Supersoft silicone
rubber to make the
casting of the MEG



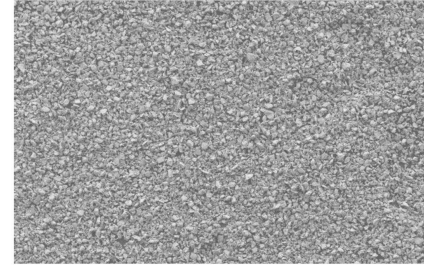
**Composed of
1 Part A**



**and 1 Part B
for 1:1 Ratio**

**Magnetic
Powder**

MQFP Powders (D50=5microns)



Signal A = MPSE EHT = 5.00 kV 20µm* Date: 30 Oct 2008
Mag = 500 X WD = 8 mm Vacuum Mode = High Vacuum Time: 13:33:02

Materials



3D Printed Mold

Mold will be a rectangle shape in which Ecoflex & Magnetic powder will be poured into to create MEG



Shirt w/ MEG

MEG will be sewed onto the inside of the compression shirt

05

Testing Protocol & Data

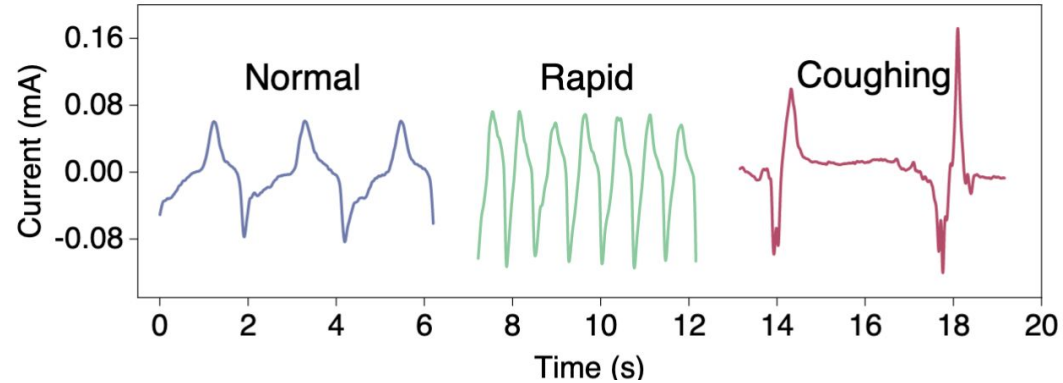


Testing Protocol

- -Wear MEG shirt and connect MEG to ammeter
- -User performs activity (running, walking, etc.)
- -Current is measured using ammeter and data is saved to a computer or phone
- -Analyze data and determine the correlation between the amount of current generated to breathing type (slow, normal, rapid)

Predicted Data

- The faster the breathing, the more peaks there are
- Similar to Lab 6: MEGs for respiration monitoring



An illustration featuring two runners, a man on the left and a woman on the right, both in motion. The man is wearing a light blue tank top and green shorts, while the woman is wearing a yellow tank top, blue shorts, and a pink cap. They are positioned on either side of a large, light blue, rounded rectangular bubble. Inside this bubble, the text '06 Challenges' is written in a bold, dark blue font. The background is white, decorated with stylized red and yellow organic shapes at the corners, several light blue maple leaves, and small green and blue circular shapes near the central bubble.

06

Challenges

Challenges	Solutions
Incorporating MEG into shirt ○ Sewing may be difficult ○ Could possibly glue	- Look into MEG textiles - May need to change thickness
Wire Placement	- Run down the seams
Damp Environment ○ Sweat can change the performance of the device	- Add fabric to sandwich between the shirt
Not Enough Energy ○ We may not produce enough mechanical energy to get efficient results	- Add more rotations on the coil - Change MEG Design



06

Project Timeline

Timeline



**Design and
produce MEG**

Week 7/8

**Incorporate
Compression
Shirt and Test**

Week 8/9

**Test device,
make
changes and
finalize**

Week 9/10

07

Future Applications





RESEARCH EXTENSIONS

Signal Processing: allowing data to be monitored remotely on a user's phone

Design and Materials: new compositions, nanostructures, and fabrication techniques → improved energy conversion capabilities

Hybrid energy harvesting: improved performance and versatility

APPLICATIONS

Health: heart rate, respiratory rate, body temperature, and activity levels

Sports Performance

Virtual Reality and Gaming: providing haptic feedback based on the wearer's movements, allowing users to feel sensations corresponding to actions within the virtual environment.

References

- Chen, Jun. “BE167LWinter2024 Lecture 4.” Winter 2024, Los Angeles, CA.
- Chen, Jun. “BE167L-Protocol for Lab 4.” Winter 2024, Los Angeles, CA.

The background features a large, light blue organic blob in the center. Surrounding it are various other shapes: a red shape in the top-left, a yellow shape in the bottom-right, a teal shape in the bottom-left, and a dark blue shape in the top-right. There are also several light blue starburst or leaf-like shapes scattered around the edges.

THANK YOU

Any Questions???

ALTERNATIVE RESOURCES

